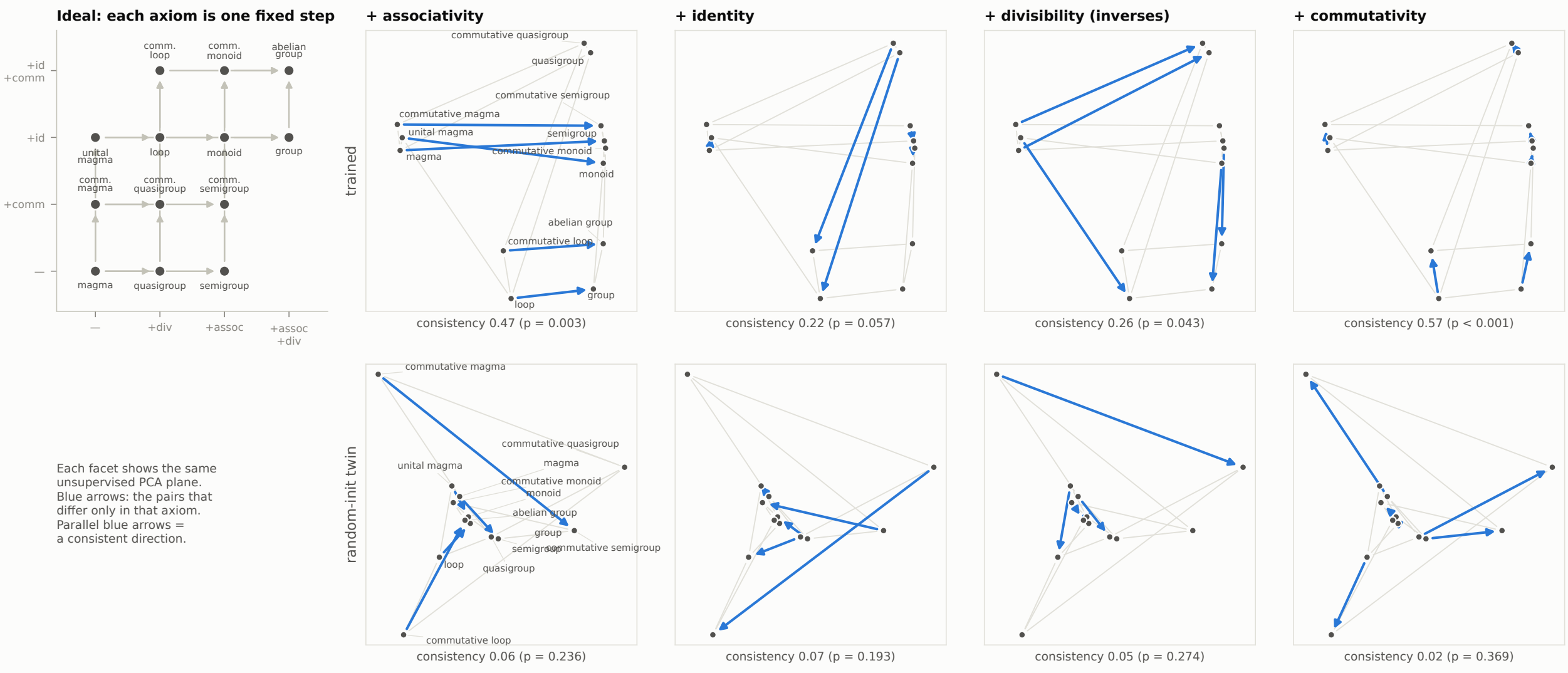
Ideal: each axiom is one fixed step
+ associativity
+ identity
+ divisibility (inverses)
+ commutativity
trained
random-init twin
comm. loop
comm. monoid
abelian group
unital magma
loop
monoid
group
comm. magma
comm. quasigroup
comm. semigroup
magma
quasigroup
semigroup
+id +comm
+id
+comm
—
—
+div
+assoc
+assoc +div
commutative quasigroup
quasigroup
commutative semigroup
commutative magma
unital magma
semigroup
magma
commutative monoid
monoid
abelian group
commutative loop
group
loop
consistency 0.47 (p = 0.003)
consistency 0.22 (p = 0.057)
consistency 0.26 (p = 0.043)
consistency 0.57 (p < 0.001)
consistency 0.06 (p = 0.236)
consistency 0.07 (p = 0.193)
consistency 0.05 (p = 0.274)
consistency 0.02 (p = 0.369)
Each facet shows the same unsupervised PCA plane. Blue arrows: the pairs that differ only in that axiom. Parallel blue arrows = a consistent direction.